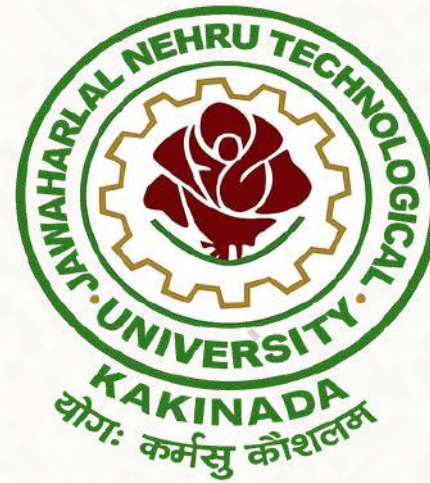


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CareerAlign AI: Intelligent Resume Shortlist Predictor and Upskilling Roadmap Generator



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Abstract

- Recruitment processes often lack transparency, as candidates do not clearly understand why their resumes are shortlisted or rejected.
- CareerAlign AI is an Artificial Intelligence-based recruitment assistance system designed to address this issue.
- The system analyzes three key inputs:
 1. Candidate Resume
 2. Job Description (JD)
 3. Predefined Eligibility Criteria
- It predicts the likelihood of a resume being shortlisted using intelligent analysis.

Abstract Contd.

- The system uses Natural Language Processing (NLP) and semantic matching to extract meaningful information from resumes.
- It performs eligibility validation based on CGPA, degree, and experience.
- Using vector embeddings and cosine similarity, it evaluates how well a resume matches a job description beyond keyword matching.
- The system generates:
 1. Shortlist prediction score
 2. Skill gap analysis (missing skills)
 3. Personalized learning roadmap using LLMs
- This solution improves recruitment transparency, enhances candidate preparation, and supports data-driven career development.

Problem Statement

The current recruitment process lacks transparency and relies heavily on keyword-based matching, which fails to capture the true relevance of a candidate's profile. Existing systems also do not provide meaningful feedback such as skill gap identification or guidance for improvement, leading to inefficiencies in hiring and uncertainty for candidates.

Motivation & Objectives

Motivation

- Increasing competition makes it difficult for candidates to understand why they are not shortlisted
- Lack of feedback in existing systems leads to confusion and no clear improvement direction
- Need for intelligent systems to provide skill gap insights and career guidance using AI and NLP

Objectives

- Develop an AI-based system for resume–job description matching
- Predict shortlist probability using semantic similarity techniques
- Perform eligibility validation based on degree, CGPA, and experience
- Identify skill gaps by comparing candidate skills with job requirements
- Provide personalized recommendations and a learning roadmap

Literature Survey

Paper	Method	Key Contribution	Limitation
Automated Resume Screener (2022)	NLP + ML based screening	Filters resumes based on skills and job requirements	No scoring system, no semantic understanding, no feedback or roadmap
Semantic Resume- Job Matching System (2026)	BERT or Transformer embeddings + NLP	Performs semantic matching using contextual understanding, improves accuracy over keyword matching	Limited to semantic matching without eligibility filtering or skill gap analysis
ProspectCV (2024)	LLM (Gemini) + NLP	Provides compatibility score, skill gap analysis, and improvement suggestions	No rule-based eligibility, no structured prediction model, heavy reliance on LLM

Research Gap:

- Existing systems focus on individual aspects such as filtering, matching, or feedback.
- However, no system integrates:
Eligibility + Semantic Matching + Prediction + Skill Gap Analysis + LLM-based Guidance

Proposed System

- The proposed system, CareerAlign AI, is an intelligent recruitment assistance platform designed to evaluate candidate suitability for a given job role.
- The system accepts a candidate's resume and job description as input and processes unstructured data using Natural Language Processing (NLP) techniques. It performs eligibility validation based on predefined criteria such as CGPA, degree, and experience. Semantic similarity techniques (TF-IDF / embeddings with cosine similarity) are used to measure the relevance between the resume and the job description. The system generates a shortlist prediction score along with a detailed skill gap analysis highlighting missing competencies.
- Additionally, it uses Large Language Models (LLMs) to provide personalized resume improvement suggestions and a structured learning roadmap. This integrated approach ensures both accurate prediction and actionable feedback for candidates.

System Requirements

Hardware Requirements:

- Basic system with internet access (Laptop/Desktop)
- Minimum 4 GB RAM (for browser usage)
- No high-end hardware required (processing done on cloud via Google Colab)

Software Requirements:

- **Development & Execution Environment:**

Google Colab – Used for developing and executing the backend ATS pipeline
Python 3.x environment (via Colab runtime)

- **Frontend:**

Gradio – Used to build an interactive user interface for resume input and result visualization

- **Backend:**

Python (Colab environment) – Handles end-to-end ATS pipeline execution including preprocessing, scoring, and decision making

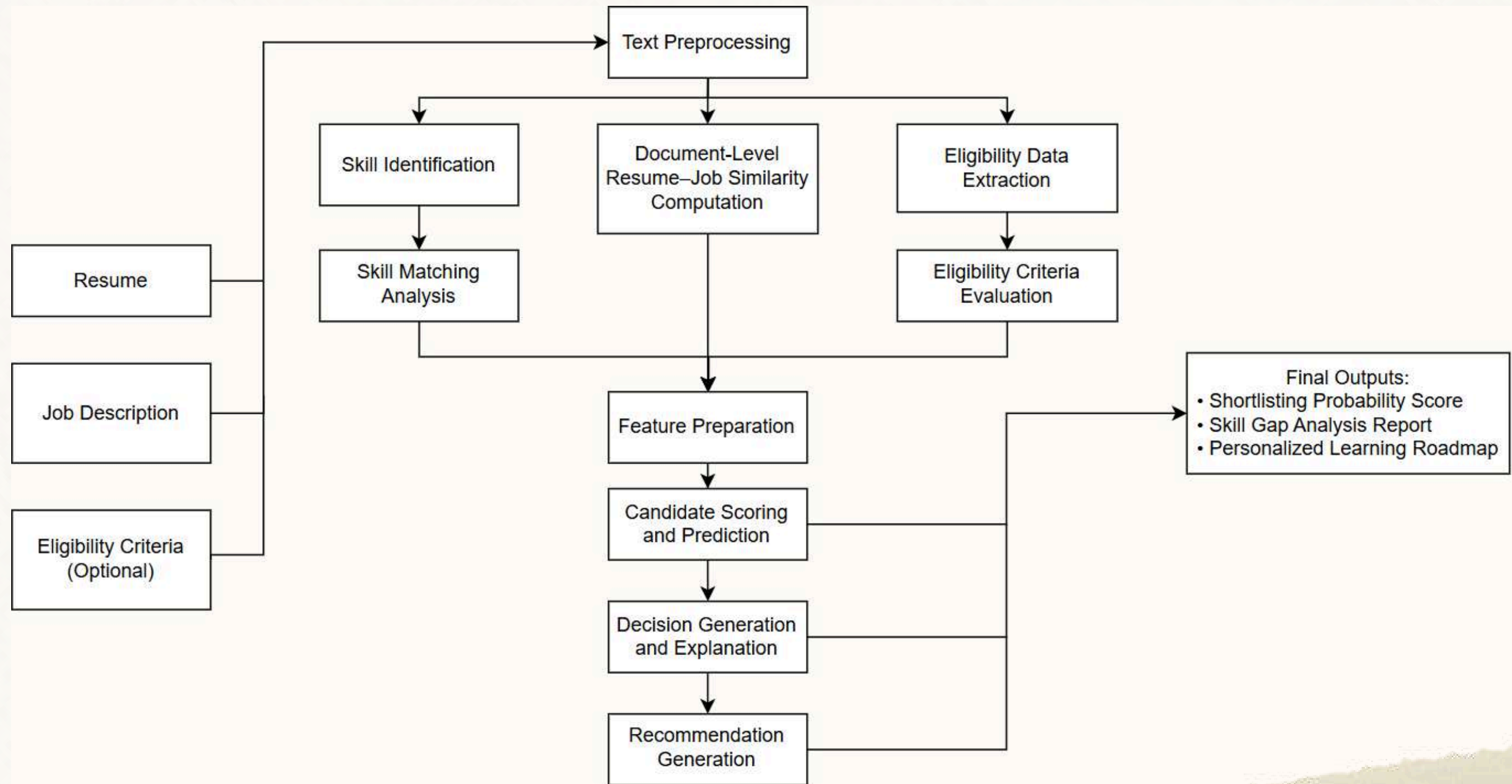
- **Natural Language Processing:**

spaCy – Used for text preprocessing, tokenization, and lemmatization

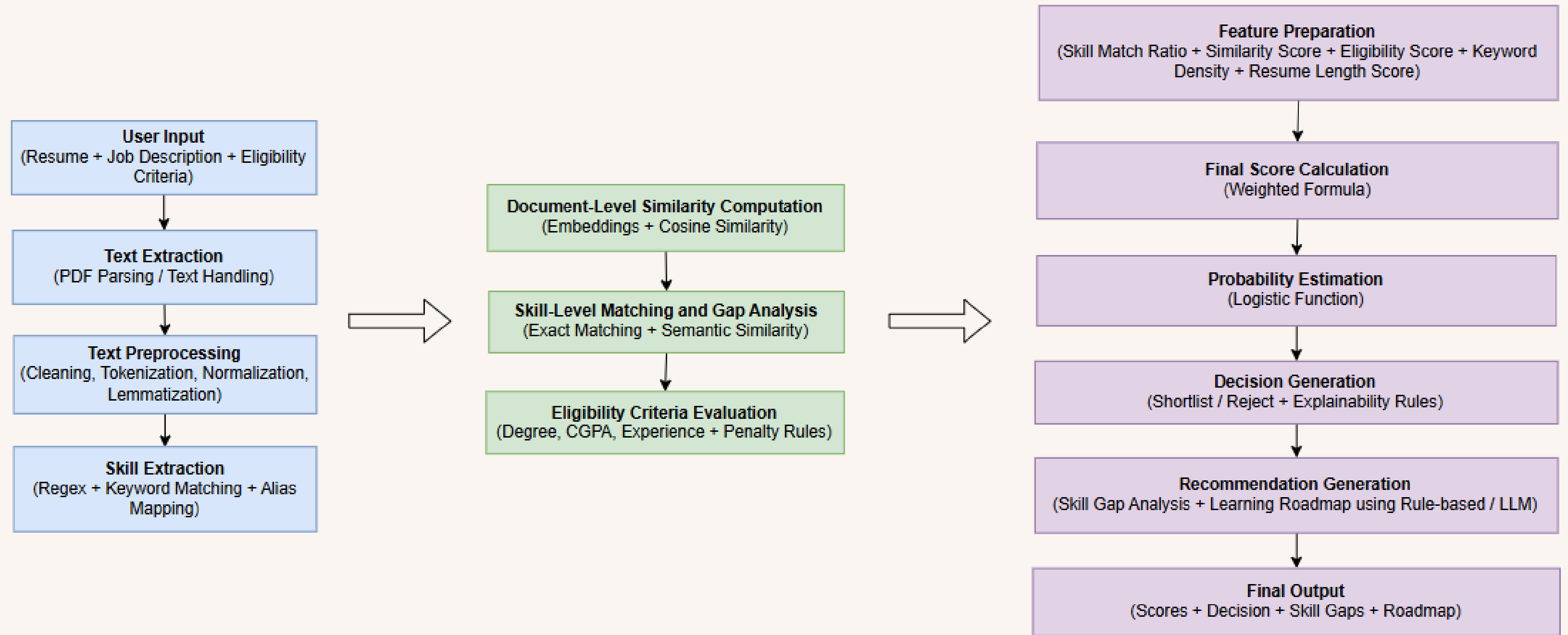
System Requirements Contd.

- **Embeddings:**
Sentence Transformers – Converts text into vector embeddings for semantic similarity
- **Similarity Measure:**
Cosine Similarity – Measures similarity between resume and job description
- **Feature Extraction:**
Custom feature extraction (similarity score, skill match, keyword density, resume length score)
- **File Handling:**
PyMuPDF (fitz) – Extracts text from PDF resumes
- **LLM Integration:**
Groq API / HuggingFace API – Generates personalized learning roadmap and recommendations
Rule-based fallback if API is unavailable
- **Supporting Libraries:**
NumPy – Numerical computations
re (Regular Expressions) – Skill extraction & parsing
math – Logistic function (probability calculation)

System Architecture



Workflow / Methodology



System Components

- **Text Preprocessing Module:** Extracts resume text from PDF and job description input using PyMuPDF. Performs cleaning, normalization, and lemmatization using spaCy to make text suitable for NLP processing.
- **Skill Extraction Module:** Identifies technical skills from resume and job description using regex-based matching. Uses skill normalization and alias mapping to handle variations like “ML” → “Machine Learning”.
- **Semantic Similarity Module:** Transforms resume and job description into vector embeddings using Sentence Transformers. Computes cosine similarity to capture contextual relevance beyond simple keyword matching.

Formula Used:

$$S_{\text{sim}} = \frac{A \cdot B}{\|A\| \times \|B\|}$$

System Components Contd.

Where:

- A = Resume embedding
- B = Job Description embedding

Logic:

- Converts text → embeddings
 - Measures similarity between vectors
 - Output scaled to 0–100
- **Skill Matching Module:** Matches resume skills with job requirements using both exact and semantic similarity. Applies weighted scoring based on importance of skills (required vs optional) to compute match strength.

Formula Used:

$$S_{\text{skill}} = \frac{\sum_{s \in M} w_s}{\sum_{s \in J} w_s}$$

System Components Contd.

Where:

- $M \rightarrow$ Set of matched skills
- $J \rightarrow$ Set of job description skills
- $w_s \rightarrow$ Weight of skill (required = 1.0, optional ≈ 0.7)

Logic:

- Required skills $\rightarrow$ weight = 1.0
- Optional skills $\rightarrow$ weight ≈ 0.7
- Semantic similarity used for partial matches

- **Eligibility Evaluation Module:** Extracts candidate details like degree, CGPA, and experience using pattern matching. Applies a soft penalty mechanism instead of strict rejection to adjust scores based on eligibility gaps.

Formula Used:

$$P_{\text{elig}} = P_d \times P_c \times P_e$$

Combined penalty factor based on eligibility criteria

System Components Contd.

Degree Penalty:

$$P_d = \max(0.5, 0.8^{\Delta_d})$$

CGPA Penalty:

$$P_c = \max\left(0.7, 1 - \frac{C_{\text{req}} - C_{\text{res}}}{10}\right)$$

Experience Penalty:

$$P_e = \max\left(0.7, 1 - \frac{E_{\text{req}} - E_{\text{res}}}{E_{\text{req}}} \times 0.3\right)$$

Where:

- $\Delta_d \rightarrow$ Degree level difference
- $C_{\text{req}}, C_{\text{res}} \rightarrow$ Required and candidate CGPA
- $E_{\text{req}}, E_{\text{res}} \rightarrow$ Required and candidate experience

Logic (Penalty Model):

- No hard rejection
- Penalty applied for missing criteria
- Ensures fairness in evaluation

System Components Contd.

- **Feature Engineering Module:** Generates features such as keyword density, and resume length score. These features represent different aspects of candidate-job alignment for scoring.

Formulas Used:

$$S_{\text{key}} = \frac{|K_{jd} \cap K_{res}|}{|K_{jd}|}$$

$$S_{\text{len}} = \begin{cases} 0.2, & \text{if } wc < 100 \\ 0.6, & \text{if } 100 \leq wc \leq 400 \\ 1.0, & \text{if } 400 < wc \leq 800 \\ 0.8, & \text{if } 800 < wc \leq 1200 \\ 0.5, & \text{if } wc > 1200 \end{cases}$$

Where:

- K_{jd} → keywords from job description
- K_{res} → keywords from resume
- wc → word count of resume

System Components Contd.

- **Scoring & Prediction Module:** Calculates final score using a weighted combination of all engineered features. Applies a logistic function to estimate shortlist probability for better decision interpretation.

Final Scoring Formula:

$$S_{\text{final}} = (0.40 S_{\text{sim}} + 0.45 S_{\text{skill}} + 0.10 S_{\text{key}} + 0.05 S_{\text{len}}) \times P_{\text{elig}}$$

Probability Formula:

$$P_{\text{shortlist}} = \frac{1}{1 + e^{-z}} \quad z = \frac{S_{\text{final}} - 50}{12}$$

Where:

- S_{sim} → Semantic similarity score
- S_{skill} → Skill match ratio
- S_{key} → Keyword density
- S_{len} → Resume length score
- P_{elig} → Eligibility penalty factor

System Components Contd.

- **Decision & Explainability Module:** Determines final outcome (Shortlist/Reject) based on computed score threshold. Provides detailed explanation including weak areas, eligibility issues, and missing core skills.

$$\text{Decision} = \begin{cases} \text{Shortlist,} & \text{if } S_{\text{final}} \geq T \\ \text{Reject,} & \text{if } S_{\text{final}} < T \end{cases}$$

Where:

- $T \rightarrow$ Decision threshold

Logic:

- Decision based on score threshold
 - Highlights weak features
 - Lists missing critical skills
- **Recommendation Module:** Generates personalized learning roadmap based on missing skills and job requirements. Uses LLM APIs (Groq/HuggingFace) or rule-based fallback to suggest courses, resources, and timelines.

System Components Contd.

Models:

- LLaMA 3.3 70B Versatile (via Groq API)
- Mistral 7B Instruct v0.2 (via HuggingFace)
- Rule-based mapping for fallback recommendations

Logic:

- Uses missing skills as input
- Generates structured roadmap
- Provides actionable improvement steps

Implementation Overview

- Developed and tested the **backend ATS pipeline** using **Google Colab**
- Implemented and validated core processing and scoring modules
- Tested system using multiple resume–job description inputs and verified outputs (scores, decision, skill gaps)
- Built an **interactive user interface** using **Gradio**
- Enabled user inputs for resume (PDF/text) and job description
- Integrated backend ATS pipeline with Gradio UI by connecting user inputs to the processing function and rendering outputs dynamically
- Achieved complete end-to-end flow: **Input → Processing → Output**
- System generates:
 - Similarity score
 - Skill match score
 - Final score & probability
 - Decision (Shortlist/Reject)
 - Missing skills and learning roadmap

Outputs - System Interface

The screenshot displays the CareerAlign AI web interface. At the top left is the logo and a brief description: 'Instant ATS resume scoring, skill-gap analysis, and a personalised learning roadmap — built for professional recruiters.' At the top right, a green status indicator shows 'SYSTEM ONLINE' and 'ATS Engine v2.0'. Below this is a 'RESUME INPUT' section with two tabs: 'Upload PDF' (selected) and 'Paste Text'. The 'Upload PDF' tab contains a large box with the text 'Drop your resume PDF here', an upload icon, and the instructions 'Drop File Here - or - Click to Upload'. To the right of this is a 'JOB DESCRIPTION' section with a text area labeled 'Paste the complete job description here...'. Below these sections is an 'ELIGIBILITY CRITERIA (OPTIONAL)' section with a text input field containing the example text 'e.g. Must have 2+ years Python experience, Bachelor's degree in Computer Science'. At the bottom center is a prominent blue button labeled 'Analyse Resume'.

- User uploads resume (PDF/text) and enters job description
- Optional eligibility criteria can also be provided
- System processes input and generates evaluation

Outputs - System Interface

 ANALYSIS EXPLANATION

 SKILL MATCH ANALYSIS MATCHED VS GAP COMPETENCIES

 MATCHED SKILLS

Matched skills will appear here after analysis...

 SKILL GAPS

Missing skills will appear here after analysis...

 FEATURE SCORES GRANULAR BREAKDOWN FROM THE ML PIPELINE

FEATURE SCORE DETAIL

Feature scores will appear here after analysis...

 LEARNING ROADMAP PERSONALISED UPSKILLING RECOMMENDATIONS

Outputs – Shortlist Case (Accepted)

**CareerAlign AI**
Instant ATS resume scoring, skill-gap analysis, and a personalised learning roadmap — built for professional recruiters.

● SYSTEM ONLINE
ATS Engine v2.0

RESUME INPUT UPLOAD PDF OR PASTE TEXT

Upload PDF

Paste Text

RESUME TEXT

Objective:
Machine Learning enthusiast with hands-on experience in Python, Machine Learning, Deep Learning, NLP, and Data Analysis.

Education:
B.Tech in Computer Science Engineering
University College of Engineering, Kakinada
CGPA: 9.1

Skills:
Python, Machine Learning, Deep Learning, Natural Language Processing, NLP, Data Analysis, SQL, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, FastAPI, AWS, Docker

Experience:
Machine Learning Intern – AI Tech Solutions (1 year)
- Developed machine learning and deep learning models
- Worked on NLP tasks such as text classification and semantic similarity
- Performed data preprocessing and feature engineering
- Built APIs using FastAPI for model deployment

JOB DESCRIPTION

Job Title: Machine Learning Engineer Intern

Responsibilities:

- Develop machine learning and deep learning models
- Work on natural language processing (NLP) tasks such as text classification and semantic similarity
- Perform data preprocessing and feature engineering
- Build APIs for model deployment using FastAPI
- Work with cloud platforms like AWS and container tools like Docker

Required Skills:
Python, Machine Learning, Deep Learning, Natural Language Processing, NLP, Data Analysis, SQL

Preferred Skills:
TensorFlow, PyTorch, Scikit-learn, FastAPI, AWS, Docker

Eligibility Criteria:

- Minimum CGPA: 7.0
- Minimum 1 year experience in Machine Learning or related field

Experience:
1+ years preferred

Outputs – Shortlist Case (Accepted)



ATS SCORE SUMMARY KEY METRICS FROM THE ANALYSIS

Similarity

83.65

Skill Match

100.0

Final Score

70.55

Probability

0.8472

Decision

✓ Accept



VISUAL BREAKDOWN

Final Score — Gauge



Feature Breakdown



Outputs – Shortlist Case (Accepted)

ANALYSIS EXPLANATION

AI Explanation:

Here's the result in simple terms:

- Your application is a good match: 83.65% similar to what we're looking for.
- You have all the necessary skills: 100% match.
- Overall, you scored 70.55%, which is good enough to move forward.
- We're 84.72% sure you're a good fit.
- Decision: You're accepted! ✓

SKILL MATCH ANALYSIS MATCHED VS GAP COMPETENCIES

✓ MATCHED SKILLS

```
aws
classification
data analysis
data preprocessing
deep learning
docker
fastapi
feature engineering
machine learning
model deployment
natural language processing
python
pytorch
scikit-learn
semantic search
sql
tensorflow
text classification
```

✗ SKILL GAPS

Missing skills will appear here after analysis...

Outputs – Non-Shortlist Case (Rejected)

**CareerAlign AI**
Instant ATS resume scoring, skill-gap analysis, and a personalised learning roadmap — built for professional recruiters.

● SYSTEM ONLINE
ATS Engine v2.0

RESUME INPUT UPLOAD PDF OR PASTE TEXT

Upload PDF

Paste Text

RESUME TEXT

Objective:
Motivated Computer Science student with strong interest in Machine Learning and Web Development.

Education:
B.Tech in Computer Science Engineering
University College of Engineering, Kakinada
CGPA: 8.5

Skills:
Python, JavaScript, HTML, CSS, Angular, Machine Learning, NLP, SQL

Projects:
1. Resume Screening System

JOB DESCRIPTION

Job Title: Machine Learning Intern

Responsibilities:

- Develop and implement machine learning models
- Work on NLP tasks
- Preprocess and clean large datasets
- Collaborate with team on AI-based solutions

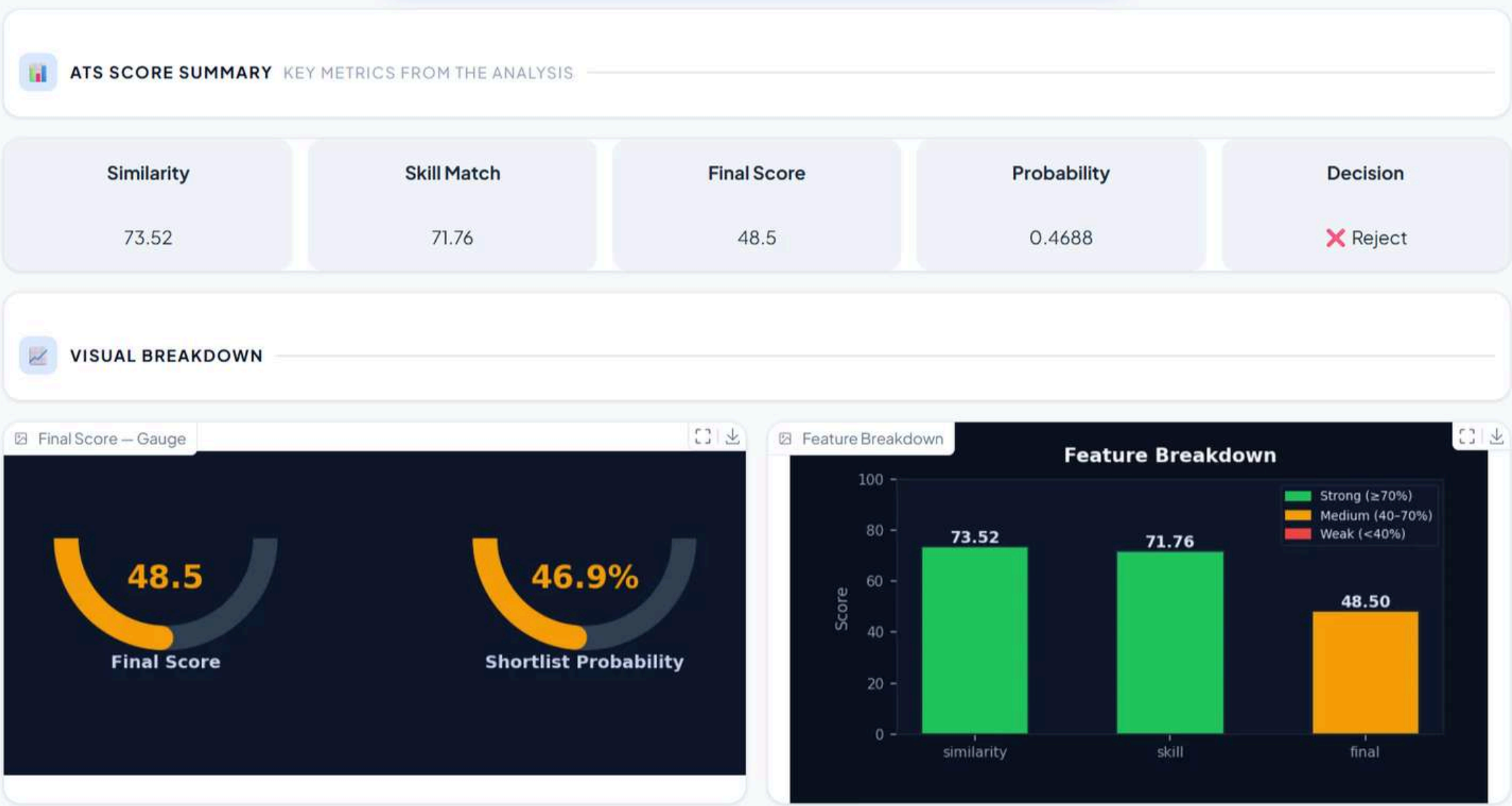
Required Skills:
Python, Machine Learning, NLP, Data Analysis, SQL

Preferred Skills:
Deep Learning, TensorFlow, PyTorch, Angular (basic)

Eligibility Criteria:

- Minimum CGPA: 7.0

Outputs – Non-Shortlist Case (Rejected)



Outputs – Non-Shortlist Case (Rejected)



ANALYSIS EXPLANATION



AI Explanation:

Here's the result in simple terms:

- Your application score is 48.5, which is not enough to move forward.
- You have some matching skills, but not all the required ones, such as:
 - Data analysis
 - Pytorch
 - Tensorflow
- Unfortunately, the decision is to reject your application.



SKILL MATCH ANALYSIS MATCHED VS GAP COMPETENCIES

✓ MATCHED SKILLS

angular
artificial intelligence
deep learning
machine learning
natural language processing
python
sql

✗ SKILL GAPS

data analysis
pytorch
tensorflow

Outputs – Non-Shortlist Case (Rejected)



LEARNING ROADMAP PERSONALISED UPSKILLING RECOMMENDATIONS

AI-Generated Roadmap (Groq):

Resume Improvement Suggestions:

- Emphasize your strong understanding of machine learning concepts and problem-solving skills, as these are key requirements for the internship.
- Highlight any relevant projects or academic work that demonstrate your proficiency in Python, NLP, and SQL.
- Consider adding a section on "Skills in Progress" or "Learning Objectives" to showcase your enthusiasm for acquiring the missing skills, such as data analysis, PyTorch, and TensorFlow.

3-Month Learning Roadmap:

Month 1: Data Analysis

- Week 1–2: Learn the basics of data analysis using Python libraries like Pandas, NumPy, and Matplotlib.
- Week 3–4: Practice data visualization and statistical analysis using datasets from Kaggle or UCI Machine Learning Repository.

Month 2: PyTorch

- Week 5–6: Learn the fundamentals of PyTorch, including tensors, autograd, and neural networks.
- Week 7–8: Implement simple PyTorch projects, such as image classification or text classification.

Month 3: TensorFlow

- Week 9–10: Learn the basics of TensorFlow, including graphs, sessions, and placeholders.
- Week 11–12: Implement projects using TensorFlow, such as building a simple neural network or using pre-trained models.

Recommended Resources:

- **Data Analysis:** + Course: Data Analysis with Python (DataCamp) + Book: Python Data Science Handbook (Jake VanderPlas) + Project: Analyze a dataset from Kaggle and create visualizations using Matplotlib and Seaborn.
- **PyTorch:** + Course: Deep Learning with PyTorch (Udemy) + Book: PyTorch Documentation (official PyTorch website) + Project: Build a simple image classifier using PyTorch and the CIFAR-10 dataset.
- **TensorFlow:** + Course: TensorFlow Tutorial (TensorFlow official website) + Book: TensorFlow for Machine Learning (Packt Publishing) + Project: Build a simple neural network using TensorFlow and the MNIST dataset.

Remember, practice is key! Work on projects and exercises to reinforce your learning, and don't hesitate to reach out to online communities or forums for help when needed. You got this!

Future Scope

- Use the proposed semantic similarity and rule-based system to generate labeled resume–job description pairs
- Create a structured dataset using system-generated scores and decisions as synthetic labels
- Train machine learning models such as Logistic Regression and Random Forest on the generated dataset
- Improve shortlist prediction accuracy using supervised learning techniques
- Enhance the system by combining rule-based scoring with trained ML models for better performance
- Further improve the user interface to provide a more seamless and interactive experience

Conclusion

- CareerAlign AI presents an intelligent approach to improve transparency in the recruitment process.
- The system integrates NLP, semantic similarity, and rule-based filtering to evaluate resume relevance effectively.
- It not only predicts shortlist probability but also provides skill gap analysis and actionable recommendations for improvement.
- The proposed solution focuses on building an explainable and scalable system that can be enhanced further using machine learning techniques.
- Overall, the project aims to support candidates in improving their job readiness and enable more efficient recruitment processes.

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Thank You!